TESTED AND HOPED ON

control software for the physical world, and the language that proves it

No mainstream language lets a team state "this loop never makes a host call, never allocates, never commands past this bound" and have the compiler prove it through every call.

So the software that steers a seven-ton flywheel is tested, reviewed and hoped on, walled behind a perimeter, and agents now write it.

WHY NOW

Coding agents write production control code this year.

WHY THE GRID

Every new inverter is a software product under UL 1741 SB, and Utah's data centers are the load.

WHY HERE

Torus is scaling a factory in Salt Lake; Hill AFB verifies flight software by review.

MENTL

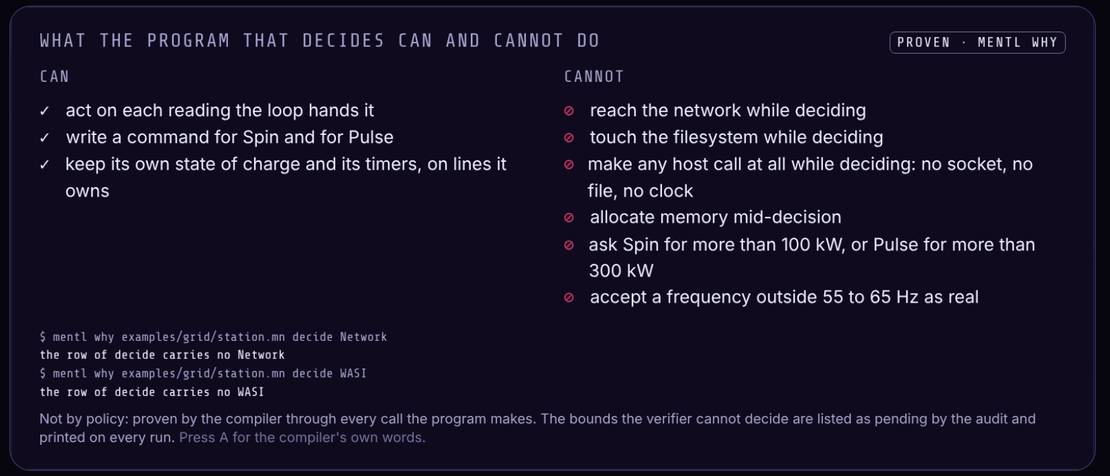
a language whose compiler proves what a program can never do

NEVER

"Never makes a host call, never allocates, never exceeds this bound": proven through every call, or the program does not build. The audit names it: Real-time safe, Sandbox.

WHY

Every fact carries its reason, walkable to the line that decided it. A dispatch explains itself, in words.



Self-hosting: the pinned compiler reproduces itself byte for byte. One WebAssembly file. Open source. 2,600 commits since March 15; first light, the compiler compiling itself to a byte-identical fixed point, on July 10.

BUILT THIS WEEK

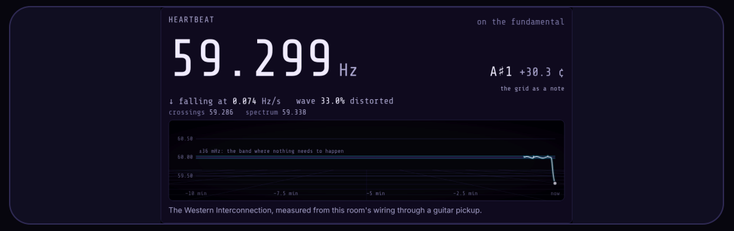
beside what existed before

BEFORE KICKOFF

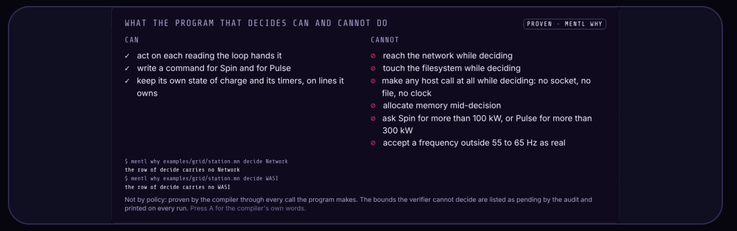
- Mentl: the compiler, the proofs, the course, the Wasatch piece made of Salt Lake's readings; 2,610 commits since March 15
- sonido: 36 effects, the plugin, the browser editor, the analysis crate, the pedal platform; 461 commits since January 25

SINCE MONDAY 4 PM

- The ear: grid frequency from a guitar, three ways, to the millihertz, on the analysis crate
- The controller in Mentl, its proofs, its refusal twin, the learner
- The page, the director, the data, the real event replayed, tomorrow's rush hour shown, the session recorded
- The gate's verdicts on proposed code, PROVEN and REFUSED with the line; the ear in the pedal







./LEDGER.SH: HUM 22 · SONIDO 3 · MENTL 9 COMMITS SINCE MONDAY 4 PM, TUESDAY'S PRINT, EACH WITH ITS TIME

WHO NEEDS IT, IN UTAH

TORUS

Their Station hears every cycle. Their security is a perimeter. The program beneath the dispatcher has never been provable, until now. The Torus judge is seeing this demo today.

HILL AFB, 309TH SOFTWARE ENGINEERING GROUP

One of the Air Force's largest software organizations, maintaining flight software for the F-16, the F-35 and the missile fleet. Assurance evidence is a deliverable, and a tool that never passes a claim it cannot decide produces it.

First conversation: Torus engineering, this month.
First hires: a controls engineer and a compiler engineer, Salt Lake City.

WHAT $10,000 UNLOCKS

ninety days, one site, two asks

Day 30

The proven controller on inverter-class hardware on my bench. The company formed; SAM.gov, SBIR.gov and DSIP registrations filed on day one. [[status]]

Day 60

One Utah site hosting an ear on a Station-class inverter, the dispatch explained to its operator: Torus's if that door opens, another's if not. The SBIR topic chosen with Nucleus Grow.

Day 90

A pilot letter from the day-60 site. Ninety days of its decisions on the page. The Phase I draft ready for AFWERX's next window.

THE $10,000

$7,000 inverter-class hardware: a three-phase inverter platform and its DC supply
$3,000 the proof gate hosted ninety days, $1,000 a month

What this room can give:
one Utah operator who will host an ear; Torus, if you can open that door
the 309th, through 47G

THE TRUSTED BASE

- The compiler is written in Mentl and the pinned binary reproduces itself byte for byte; that fixed point is the release.
- The verifier is sound for what it decides and incomplete by design: effects and types are judged by check; a bound it cannot decide is listed as pending by audit and printed on every run.
- One false proof is known and documented: a recursion deep enough to exhaust the stack traps although its function says it cannot.
- Rust prevents memory bugs, not effects; Simulink and SCADE verify models and generate C. Neither states "never reaches the network" as an obligation through every callee, proves a function allocation-free, or derives a learning rule from the chain.

RUNNING ON A CONTROLLER

- The proof is at compile time. The artifact is one WebAssembly module importing WASI and nothing else.
- It runs wherever wasmtime runs, which includes the Linux supervisors that sit beside inverters today; the demo's controller runs that way and answers each reading in under a millisecond.
- A native backend is on the roadmap; the pedal in the demo runs the ear natively on a Cortex-M7 today.
- The learner keeps learning inside a loop proven not to allocate: learning at the edge that cannot stall.
- A running computation is a value: the compiler's own resume verb restores a persisted image and continues, so durability is construction, not infrastructure.

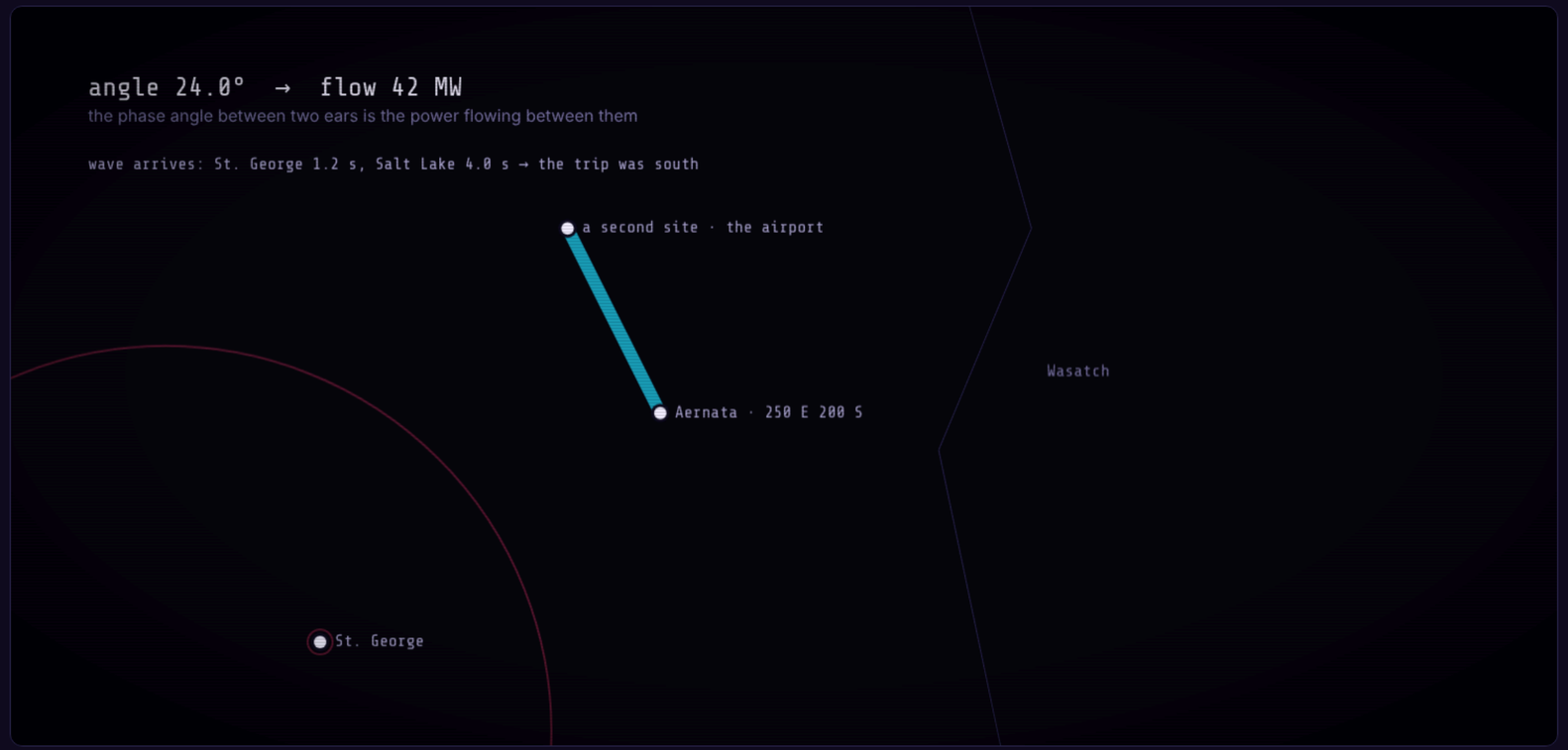
THE GRID THROUGH A DSP LIBRARY'S LENS

IN SONIDO	ON THE GRID
Pitch detection (the tuner)	Frequency: 60 Hz is a B, 49 cents flat; the 2019 event is a key change to A♯
THD (the distortion analyzer)	Power quality: the wave's timbre says what is plugged in
A compressor	Droop control: threshold, ratio, ceiling, hold; the controller's fast band is exactly that curve
A noise gate	Islanding detection
Tuning to a reference, the beat going to zero	Reconnecting an island: a synchroscope is a tuner
The correlation meter	Two sites: the angle between them is the flow
LMS (the adaptive filter)	The learner, written twice: by hand in Rust, and derived by the compiler in Mentl

THE MESH

MESH

SIMULATED



Two Stations with GPS time see the **angle** between them, which is the flow on the line between them, and an event's wave arrives at one before the other. FNET has done this from wall outlets since 2004; a fleet that both sees and answers at 10C has not existed. Simulated from the day's shape and the replayed event.